

	the direction of the magnetic flux density.
7(a)	A photon is a quantum of electromagnetic radiation, where the energy in a photon equals to the product of its frequency and Planck constant.
7(b)	When white light passes through the gas, the electrons in the gas absorbs the photons with energy values corresponding to the energy differences between the discrete energy levels in the gas. After the absorption, the excited electrons are unstable and hence will eventually fall back to the lower energy state by emitting photons of the same wavelength of photons that were initially absorbed. The emission is equally in all directions. Hence the intensity of the radiation in the direction of the incident photons is reduced. This contributes to the dark lines (whose wavelengths are that of the absorbed photons) in the continuous spectrum.

No.	Solution
	The other photons in the white light will pass through unabsorbed. Hence, the spectrum of transmitted light will consist of photons that were not absorbed by the gas, and the dark lines correspond to the photons absorbed by the gas.
7(c)(i)	$E_2 = \frac{-13.6}{2^2} = -3.40 \text{ eV}$
7(c)(ii)	$E_3 = \frac{-13.6}{3^2} = -1.51 \text{ eV}$ $\Delta E = -1.51 - (-3.40) = 1.89 \text{ eV}$ $E = \frac{hc}{\lambda}$ $\lambda = \frac{hc}{E}$ $= \frac{(6.63 \times 10^{-34})(3.0 \times 10^8)}{1.89 \times 1.60 \times 10^{-19}}$ $= 6.6 \times 10^{-7} \text{ m}$
7(c)(iii)	The energy of the highest energy level, where $n = \infty$, is zero. Since the energy of the energy level decreases with decreasing n , the energy of each energy level will be less than zero and thus negative.
8(a)	Using $v^2 = u^2 + 2as$